

Proving some of the things used for further proving

↳ A set of vectors is orthogonal $\Rightarrow$ they are linearly independent.

let $u_1, u_2, \dots, u_n$ be orthogonal set of vectors

Now let $c_1 u_1 + c_2 u_2 + \dots + c_n u_n = 0$

Inner product
with u_i

~~Multiply by c_i~~ $\Rightarrow$

$$c_i \langle u_i, u_i \rangle = 0$$

$$\Rightarrow \boxed{c_i = 0} \quad \forall i$$

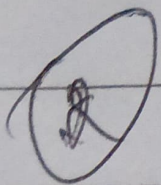
Hence

$\{u_1, u_2, \dots, u_n\}$
are linearly
independent.

Since

$$\langle u_i, u_j \rangle = 0$$

for $(i \neq j)$



↳ A is invertible iff $\det(A) \neq 0$

we know $A^{-1} = \frac{\text{adj}(A)}{\det(A)}$ therefore for A^{-1} to exist $\det(A) \neq 0$

ASS-1

(Q1)

$$v_1 = x_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$v_2 = x_2 - \left(\frac{\langle v_1, x_2 \rangle}{\langle v_1, v_1 \rangle} \right) v_1 = \begin{bmatrix} 3 \\ 4 \\ 2 \end{bmatrix} - \left(\frac{7}{2} \right) \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 - \frac{7}{2} \\ 4 - \frac{7}{2} \\ 2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ 2 \end{bmatrix}$$

Since inner product is standard inner product therefore

$$v_1 \cdot x_2 = \begin{bmatrix} 3 \\ 4 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 3 \cdot 1 + 4 \cdot 1 + 2 \cdot 0 = 7$$

$$v_1 \cdot v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 1 + 1 + 0 = 2$$

Therefore $v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ 2 \end{bmatrix}$ is the orthogonal basis for the subspace $\mathbb{R}^3$ spanned by the given vectors.

(2) a) Since AB is ~~not~~ invertible therefore $\det(AB) \neq 0$ — (1)

Also calculating dimensions of A & B

$$A \rightarrow n \times n \quad B \rightarrow m \times m$$

Since A & B are given to sq

Now since for AB to exist $\text{col}(A) = \text{rows}(B)$

$$\rightarrow A_{n \times n} \quad B_{n \times n}$$

$$n \quad n = m$$

$$\rightarrow AB_{n \times n}$$

$$\text{Hence } |AB| = |A||B| \quad \det(AB) = \det(A) \det(B)$$

$$\text{from (1)} \quad \det(A) \cdot \det(B) \neq 0$$

$$\text{therefore } \det(A) \neq 0 \quad \& \quad \det(B) \neq 0$$

Hence A is invertible & B is invertible.

② Given A, B are invertible
 To show A, B are invertible.

② a) Given $A = A^T$ & A is invertible

To prove A^{-1} is symmetric
 $\rightarrow (A^{-1})^T = A^{-1}$

Proof $AA^{-1} = I$ } A being invertible

Taking transpose

$$(AA^{-1})^T = I^T$$

$$(A^{-1})^T A^T = I$$

} By fundamental properties $(AB)^T = B^T A^T$

We know that $A^T = A$

$$2) (A^{-1})^T A = I \quad \left\{ \begin{array}{l} BA = I \end{array} \right.$$

By uniqueness of inverse $B = A^{-1}$

$$\therefore \boxed{(A^{-1})^T = A^{-1}} \quad \underline{\underline{H/P}}$$

Q3

a) A is invertible iff $|A| \neq 0$

therefore

$$\det \begin{pmatrix} K & -K & 3 \\ 0 & K+1 & 1 \\ K & -8 & K+1 \end{pmatrix} \neq 0$$

$$K(K^2-1+8) + K(-K) + 3(-K(K+1)) = \cancel{K^3-K+8} K^3 + 7K - K^2 - 3K^2 - 3K$$
$$= K^3 - 4K^2 + 4K \neq 0$$

$$K(K^2 - 4K + 4) \neq 0$$

$$K(K-2)^2 \neq 0$$

Therefore for any ~~value~~ value $K \neq 0, 2$
the given matrix is invertible

$$\Rightarrow K \in \mathbb{R} - \{0, 2\}$$

$$b) \det \begin{pmatrix} k & k & 0 \\ k^2 & 2 & k \\ 0 & k & k \end{pmatrix} = k(2k - k^2) - k(k^3) = 2k^2 - k^3 - k^4 \neq 0$$

$$\Rightarrow k^4 - k^3 - 2k^2 \neq 0$$

$$k^2(k^2 + k - 2) \neq 0$$

one root is (-1)

$$\Rightarrow k^2(k+2)(k-1) \neq 0$$

Hence for all ~~non~~ values of

$$k \neq 0, -2$$

$$k \neq 0, -2$$

$$\Rightarrow k \in \mathbb{R} - \{0, -2\}$$

Will ~~be~~ be invertible

$$1 < \frac{k^2 + 2}{k^3 - k^2 - 2} < 2$$

(4) Given $u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ $u_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ $u_3 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$

$$u_1 \cdot u_2 = 0$$

$$u_2 \cdot u_3 = 1 - 1 = 0$$

$$u_3 \cdot u_1 = 1 + 1 - 2 = 0$$

$\therefore u_1, u_2, u_3$ are orthogonal vectors

Since orthogonal vectors are ~~also~~ always linearly independent.

$\therefore u_1, u_2$ & u_3 are linearly independent.

No. of dimension of $\mathbb{R}^3 = 3$

non-zero

from previous theorems ~~basis~~ any linearly independent set of 3 vectors forms the basis for $\mathbb{R}^3$

Therefore $\{u_1, u_2, u_3\}$ form the orthogonal basis for $\mathbb{R}^3$

$$w = \begin{bmatrix} 7 \\ 9 \\ 10 \end{bmatrix} \text{ let coordinates be } x, y, z$$

$$w = xu_1 + yu_2 + zu_3$$

$$\Rightarrow \begin{bmatrix} 7 \\ 9 \\ 10 \end{bmatrix} = x \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 7 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$$

$$\Rightarrow x + y + z = 7 \quad (2)$$

$$x + y + z = 9 \quad (3) \quad x = 2z + 10 \quad (1)$$

$$x - 2z = 10$$

$$\text{Putting } x = 2z + 10 \text{ in } (2) \text{ \& } (3)$$

$$y + 3z = -3$$

$$y + 3z = -1$$

$$\Rightarrow \boxed{z = -\frac{2}{3}}$$

$$\boxed{x = \frac{26}{3}} \text{ from } (1)$$

$$\text{Putting } z \text{ in } (2)$$

$$\boxed{y = -1}$$

$$\Rightarrow \text{coordinates of } w \text{ w.r.t basis } \{u_1, u_2, u_3\} = \left\{ \frac{26}{3}, -1, -\frac{2}{3} \right\}$$

$$\boxed{(A^{-1})^{-1} = A} \quad \underline{\underline{H/P}}$$

⑤ a) By definition an orthogonal matrix is a matrix whose columns form an orthonormal set

therefore
$$v_1 = \begin{bmatrix} 1/3 \\ 1/3 \\ -1/3 \end{bmatrix} \quad v_2 = \begin{bmatrix} 1/2 \\ -1/2 \\ 0 \end{bmatrix} \quad v_3 = \begin{bmatrix} 1/5 \\ 1/5 \\ 2/5 \end{bmatrix}$$

must form an orthonormal set

↳ orthogonal or unit vectors

But
$$\|v_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 0} = \frac{1}{\sqrt{2}} \neq 1$$

therefore $\{v_1, v_2, v_3\}$ does not form an orthonormal set.

$$\text{let } Q = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

5) b) ~~We know that a matrix Q is or~~

$$v_1 = \begin{bmatrix} a \\ b \end{bmatrix} \quad v_2 = \begin{bmatrix} c \\ d \end{bmatrix}$$

$\{v_1, v_2\}$ must form an orthonormal set

$$\rightarrow a^2 + b^2 = 1 \quad (1) \quad \|v_1\| = 1$$

$$c^2 + d^2 = 1 \quad (2) \quad \|v_2\| = 1$$

$$ac + bd = 0 \quad \langle v_1, v_2 \rangle = 0$$

Case 1) $a \neq 0$

$$\boxed{c = \frac{-bd}{a}} \rightarrow \text{from eq. (2)} \quad \frac{b^2 d^2 + d^2}{a^2} = 1$$

(3)

$$\frac{d^2 (a^2 + b^2)}{a^2} = 1 \rightarrow d^2 = a^2 \because a^2 + b^2 = 1 \text{ from (1)}$$

$$d = a$$

$$\text{or } d = -a$$

$$\rightarrow \boxed{d = \pm a}$$

Putting in (3) $c = \frac{-b}{a}(a) \text{ or } \frac{-b}{a}(-a)$

$$c = -b \quad \text{or} \quad c = b$$

$$\rightarrow Q = \begin{bmatrix} a & b \\ -b & a \end{bmatrix} \quad \text{or} \quad Q = \begin{bmatrix} a & b \\ b & -a \end{bmatrix}$$



can replace
b with -b
as well
to get exact form.